Building figures with Inkscape

October 29th, 2026

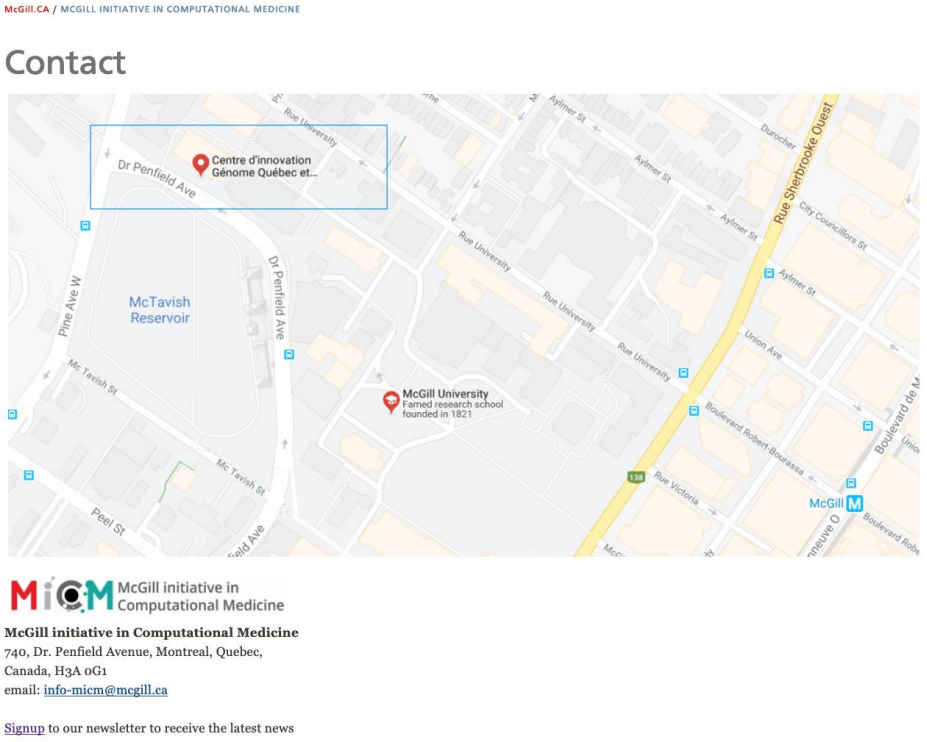
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Senthilkumar Kailasam

McGill | CDSI Computational and Data Systems Institute

C3G Canadian Centre for Computational Genomics

D2R DNA TO RNA | HeDS

Mission : aims to deliver inter-disciplinary research programs and empower the use of data in health research and health care delivery



https://www.mcgill.ca/micm

Outline:

1. Introduction to scientific figure design and Inkscape
2. Importing and editing plots
3. Building a multi-panel figure
4. Exporting and closing remarks

This is an interactive workshop :)

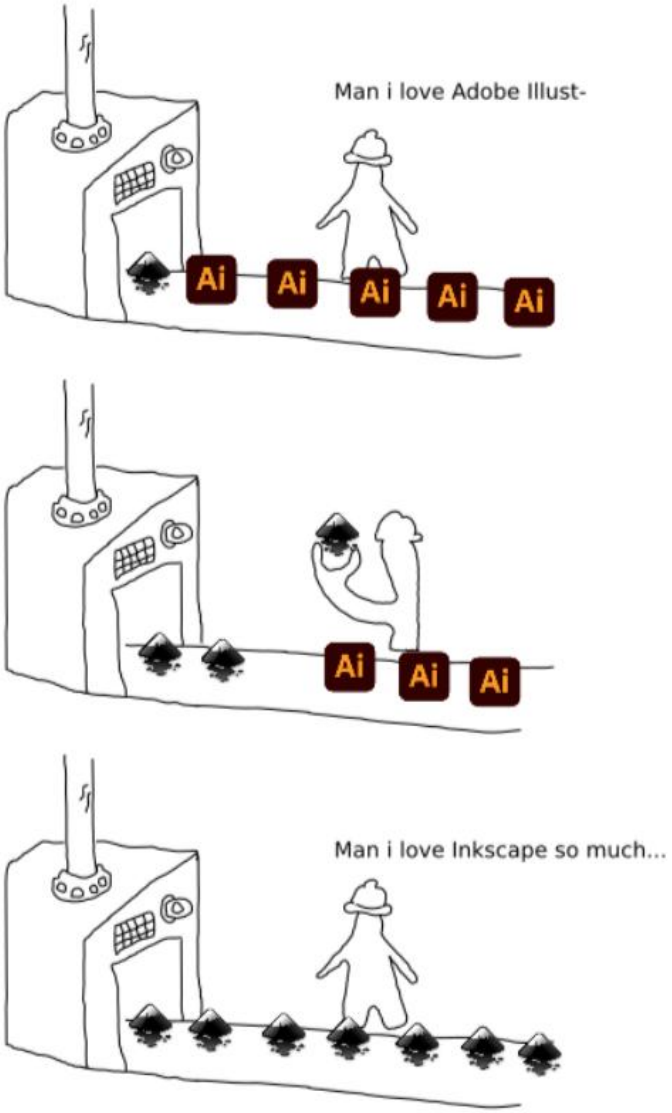
Feel free to interrupt or raise your hand to ask questions

The objective for today

To use Inkscape to transform a collection of independent plots into a polished, publication-ready figure

Part 1: Introduction to scientific figure design and Inkscape

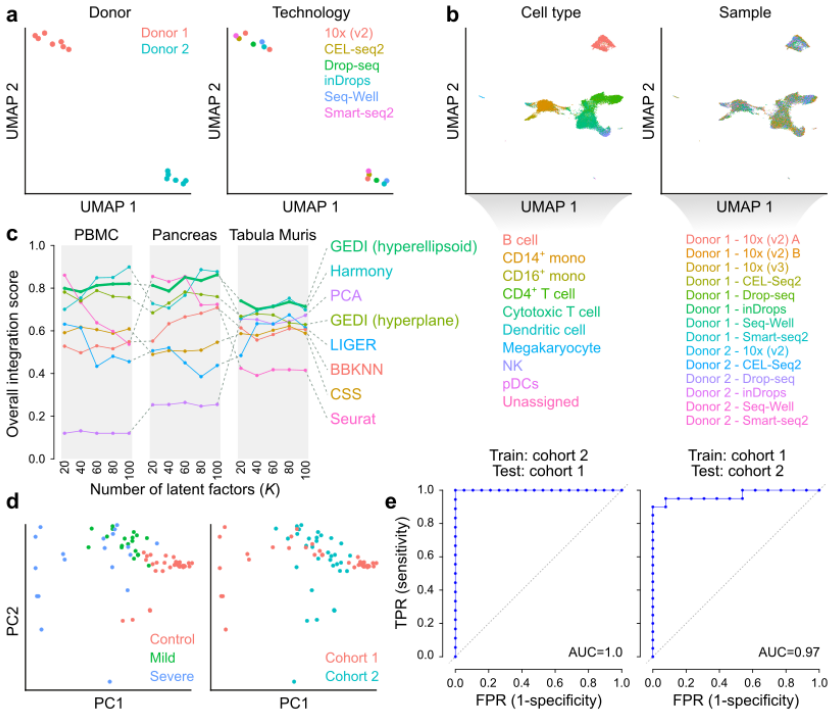
Why use Inkscape for scientific figures?



- It's free!
- Open source
- Cross-platform (Mac, Windows, GNU/Linux)
- Supports vector graphics
- Enables creation of high-resolution figures

What can you do with Inkscape?

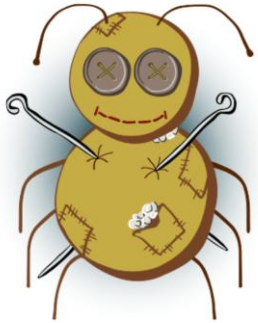
Scientific figures

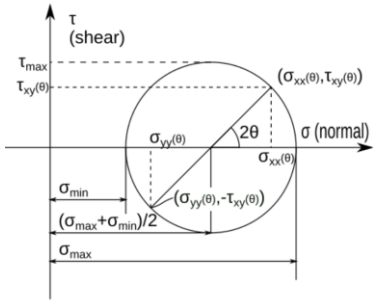


But also, much more!

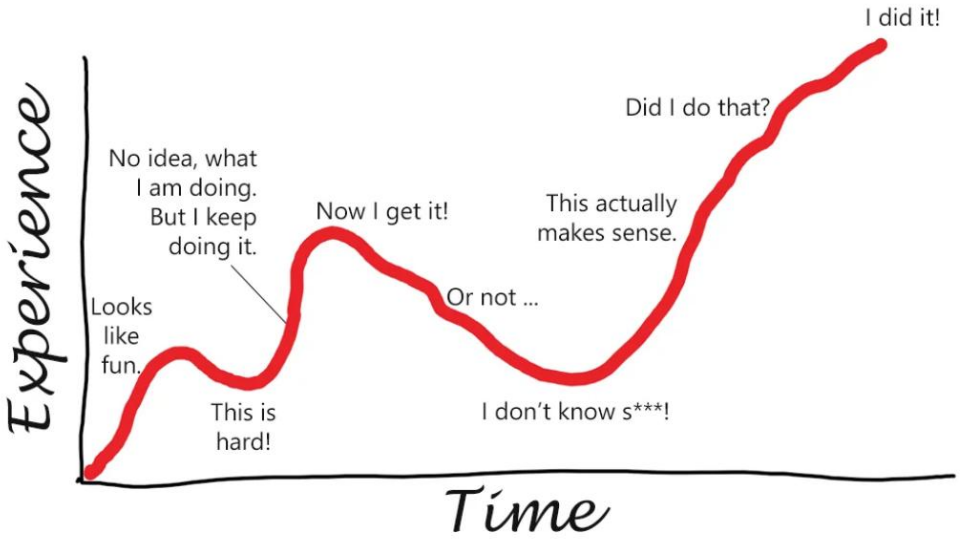








Inkscape can be hard…



But it's worth it

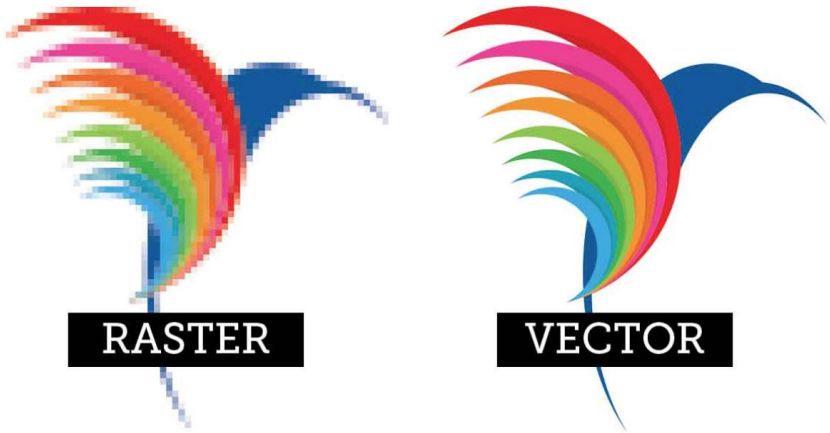
Vector vs raster graphics

Raster graphics

- Made of **pixels** arranged in a grid
- Resolution-dependent
- Become blurry or pixelated when enlarged
- Best for photographs and complex images
- Larger image dimensions are needed for high-quality printing

Vector graphics

- Made of **mathematical shapes and paths**
- Resolution-independent
- Can be scaled to any size without losing quality
- Individual elements remain editable
- Ideal for plots, diagrams, illustrations, and figure assembly

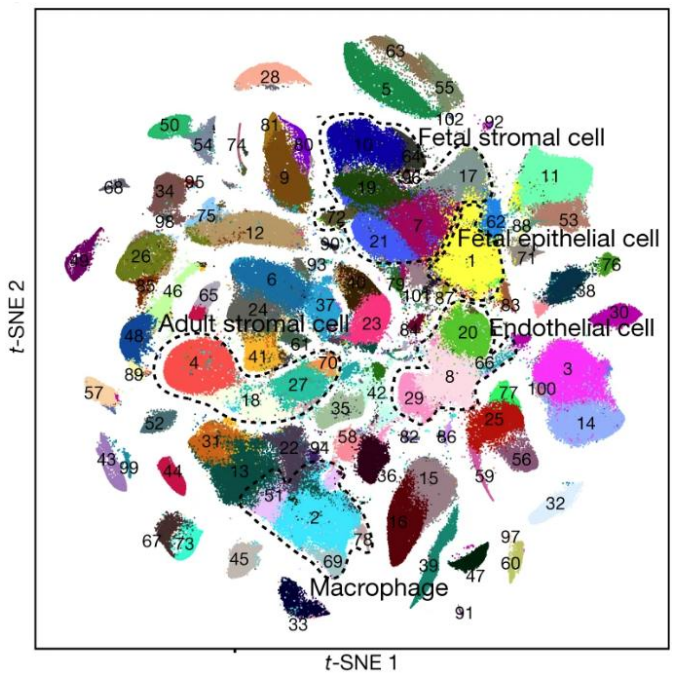


When to use vector or raster images?

Raster graphics

Complex technical images:

- Microscopy images
- Histology slides
- Plots with hundreds of thousands of points



Vector graphics

When possible, use vector-based graphics:

- Most of the computer-generated plots
- Diagrams and schematics
- Text, labels, arrows and annotations

Either:

- Use rasterization options while exporting your figures (e.g. ggrastr in R)
- Export rasterized versions of the plot without labels

Common file formats

Raster graphics

- PNG (.png)
- TIFF (.tif, .tiff)
- JPEG (.jpg, .jpeg)

Vector graphics

- SVG (.svg)
- PDF (.pdf)
- EPS (.eps)

Best practices for publication-quality figures

Use vector formats whenever possible to preserve quality and editability

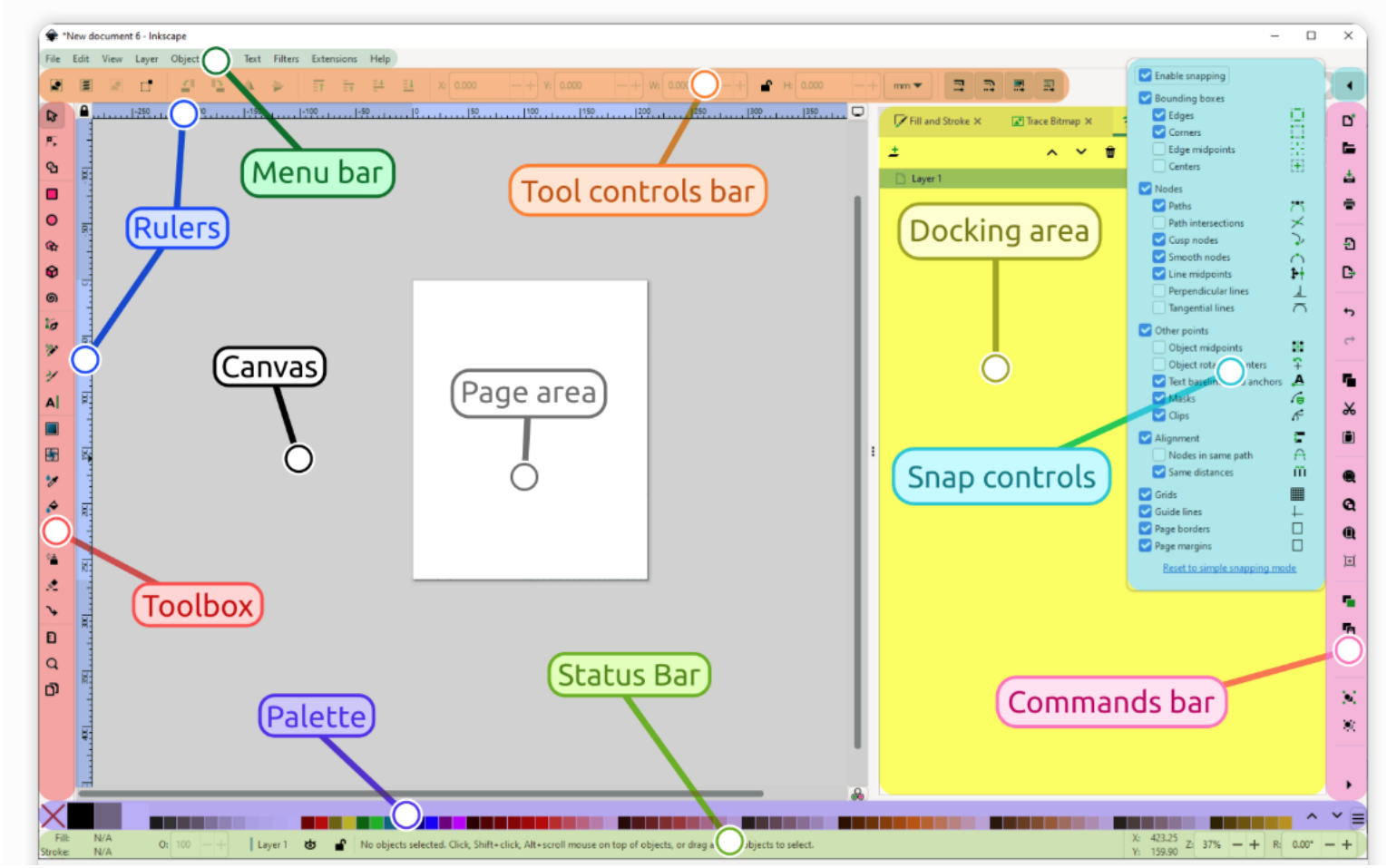
Ensure text, labels, and legends remain readable

Align panels and use consistent spacing to create a balanced layout

Use color thoughtfully and consider accessibility (e.g., colorblind-friendly palettes)

Export figures using journal-appropriate formats and dimensions

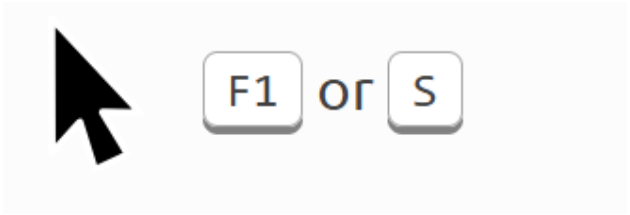
Overview of the Inkscape interface



Part 2: Importing and editing plots

Let’s familiarize with some tools

Selector tool



Zoom tool

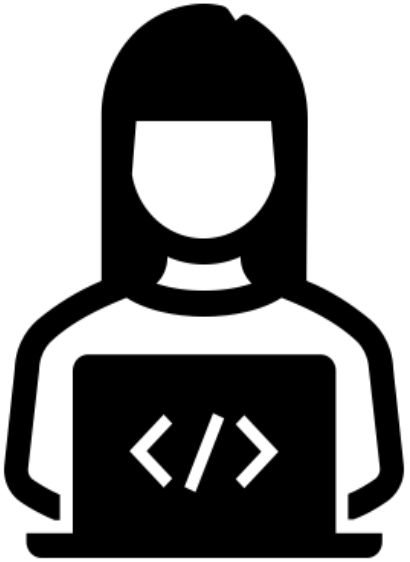


F3 or Z

Text tool

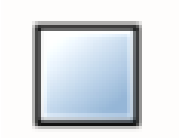


F8 or T



Let’s create some basic shapes

Make rectangles



R

Make circles, ellipses and arcs

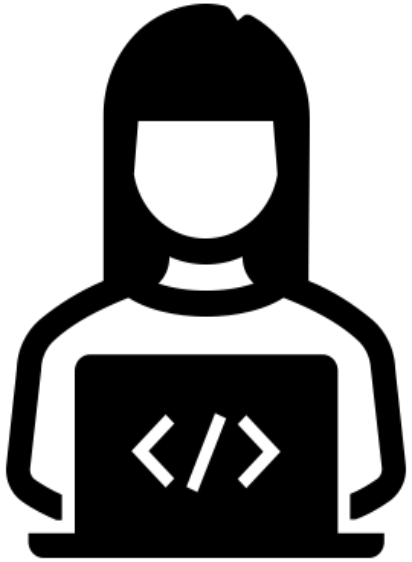


E

Make polygons, stars



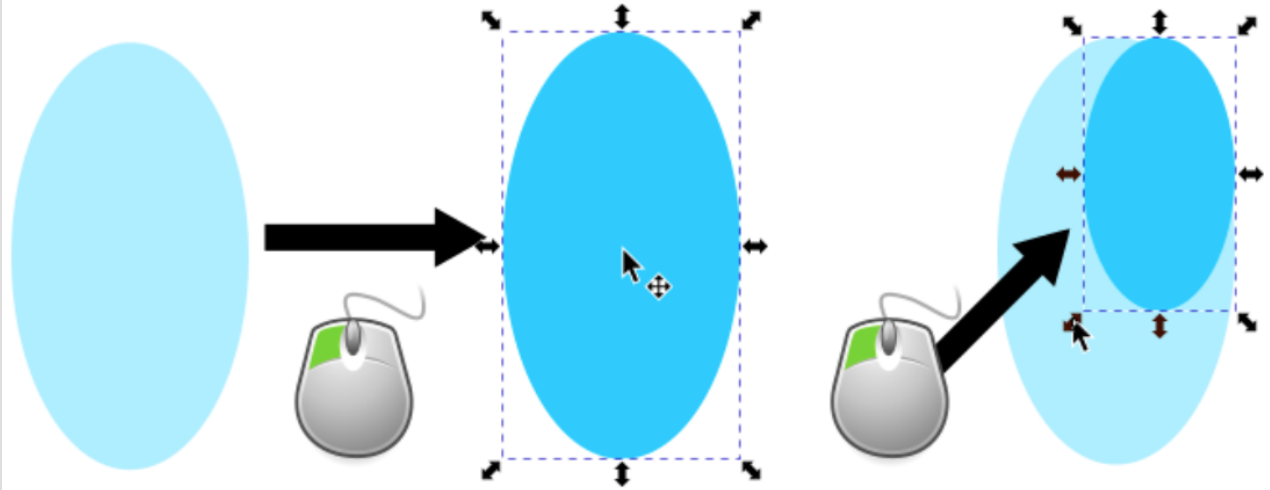
*



Moving, scaling, skewing and rotating objects

The Selector Tool

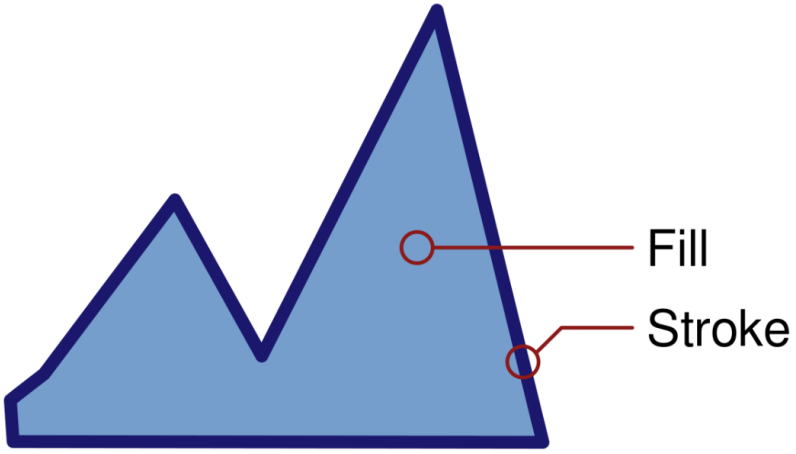
F1 or S



Drag to move the selection. Transformations (such as moving, scaling, rotating) are easy thanks to the two-way arrows.

Colors

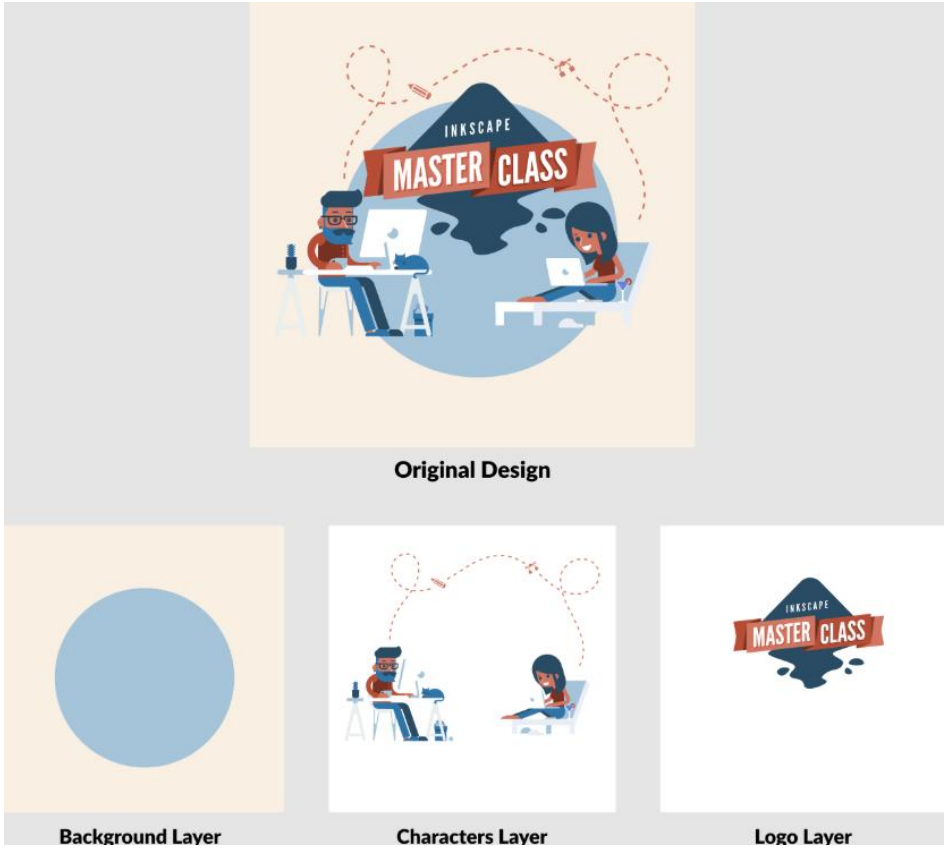
For a vector object, a separate style can be defined for its outline (or stroke) and its main color (or fill). This can be a plain color (e.g. dark red), a pattern (e.g. stripes or flowers), or a gradient (e.g. a smooth transition from green to blue).





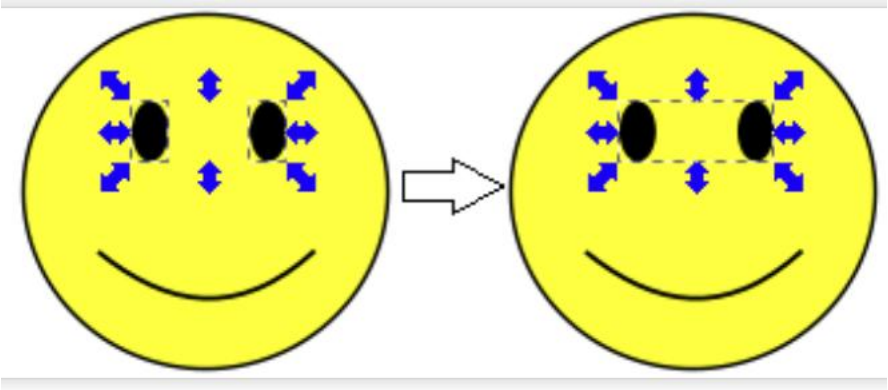
Understanding layers

Layers allow you to organize elements of your figures into sections

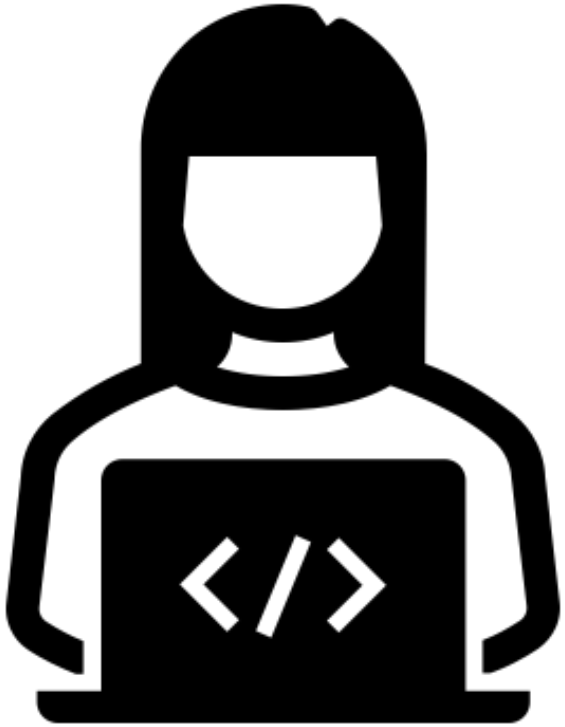


Group objects

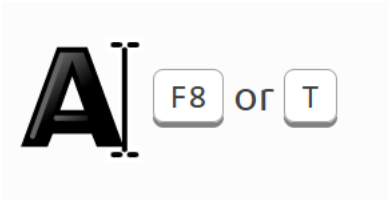
Grouping allows you to work with multiple objects as a single object: any changes you make will equally affect all members of the group, and the entire group can be copied, duplicated or cloned as a unit



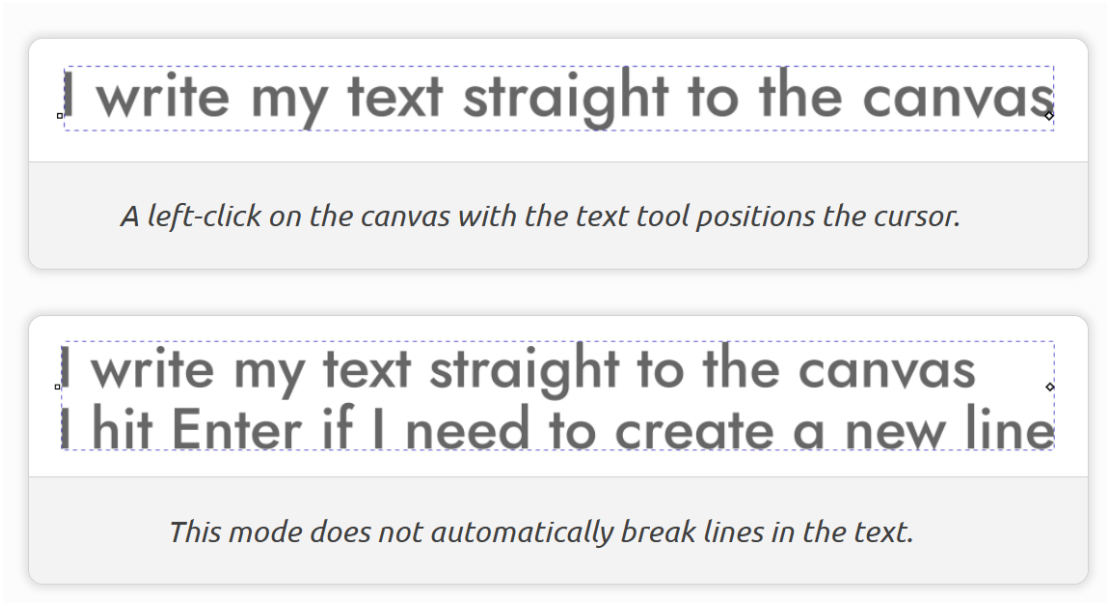
Importing images into Inkscape



Managing fonts and text elements



Two options of how to create text:

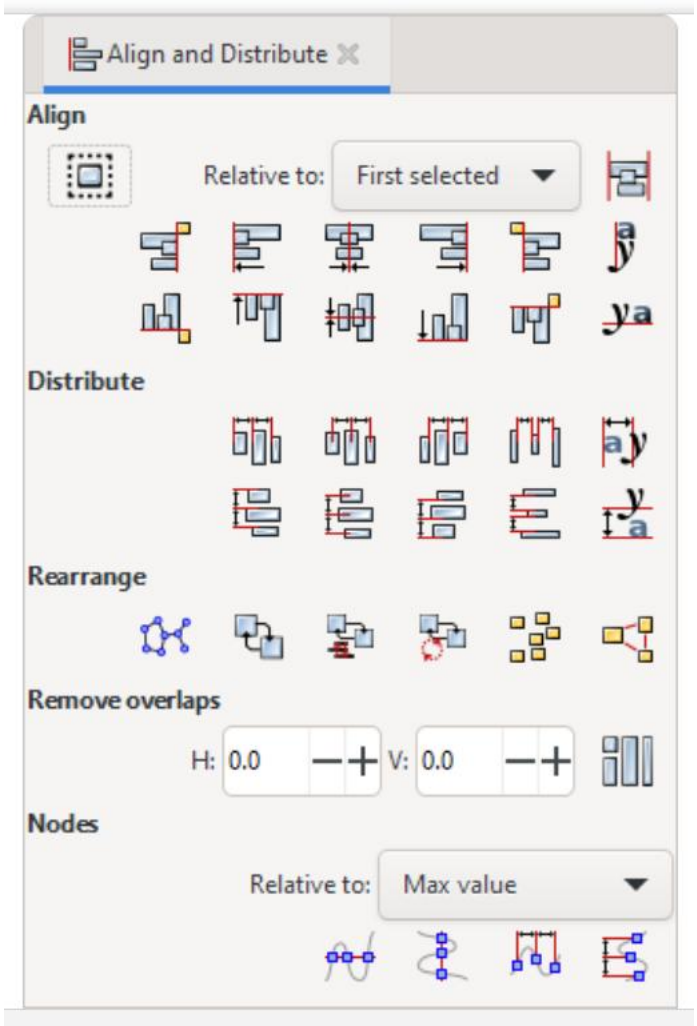


Part 3: Building a multi-panel figure

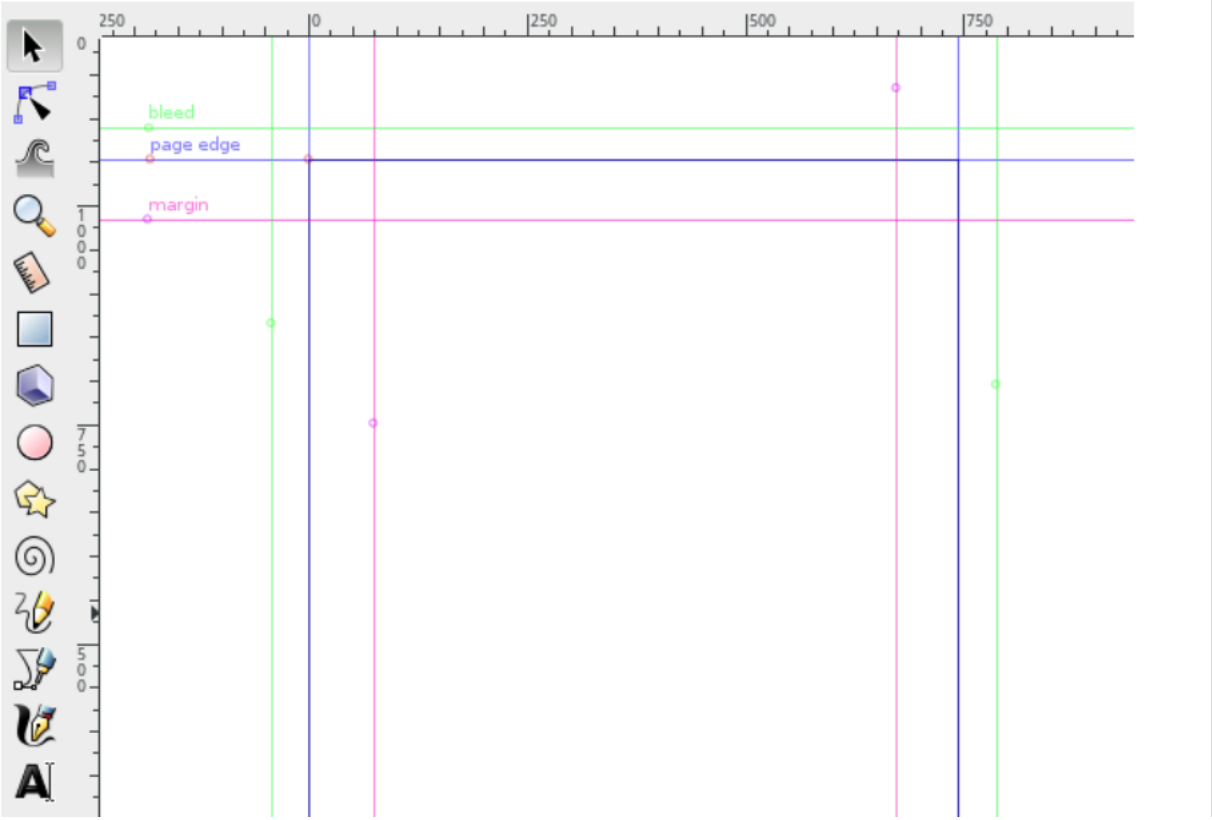
Alignment and distribution tools

Align and Distribute

Shift + Ctrl + A



Guides



Copying a color

There are several ways to apply the same style to multiple objects:

- **Selecting multiple objects**: select all the objects whose color you want to change (hold `Shift` and click, or drag a selection box around them), then open the `Fill and Stroke` dialog, select a color. It will be applied to all selected objects automatically.
- **Use the Dropper tool**: select an object, then activate the Dropper tool by clicking on its icon 🖉, then click on the canvas, on an area where it has the color that you want to apply. The selected object's color will change accordingly.
- **Copy the complete style of an object to apply it to another object:** select the object, copy it to the clipboard with `Ctrl` + `C`, then select the second object and paste only the style with `Shift` + `Ctrl` + `V`. This not only copies the fill and stroke color, but also the stroke width and style, a text's font style and any patterns or gradients.

Clipping and masking

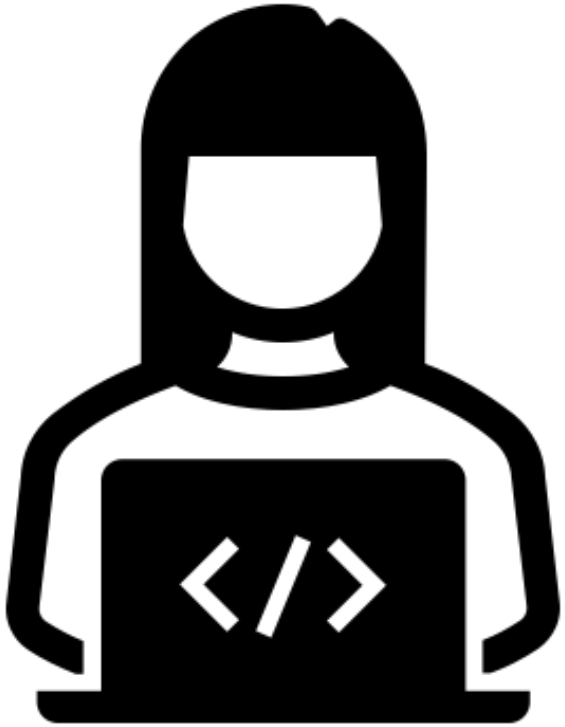
Put the object that will serve as clip (or mask) above the object that you want to clip (or mask)



Clipping

Masking

Let’s build a multi-panel figure



Part 4: Exporting and closing remarks

Saving and exporting

Always save your current work as .svg (Inkscape .svg)

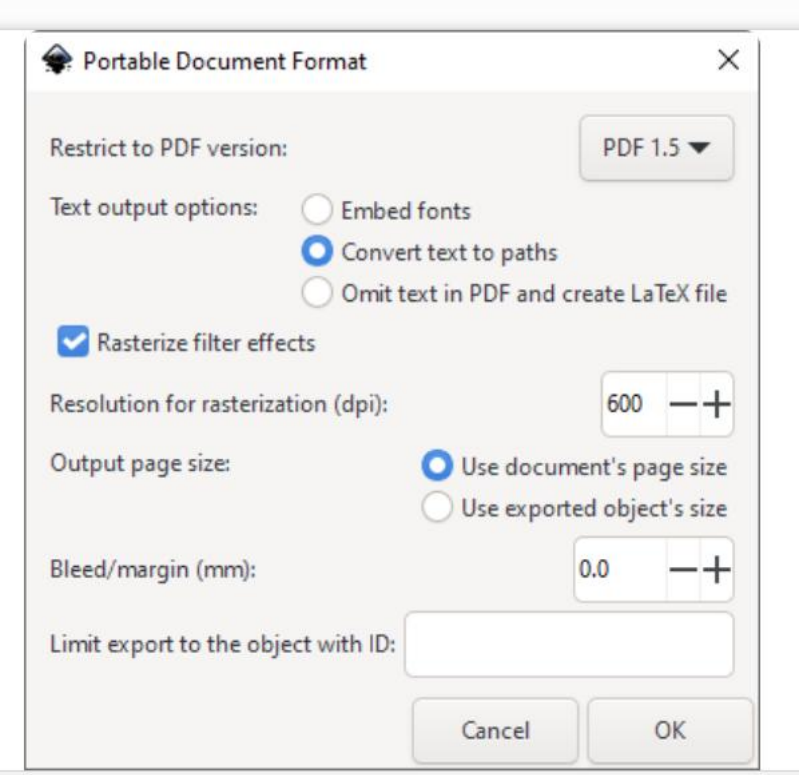
For exporting, you have several options:

- For high-quality figures (Save a copy to PDF)

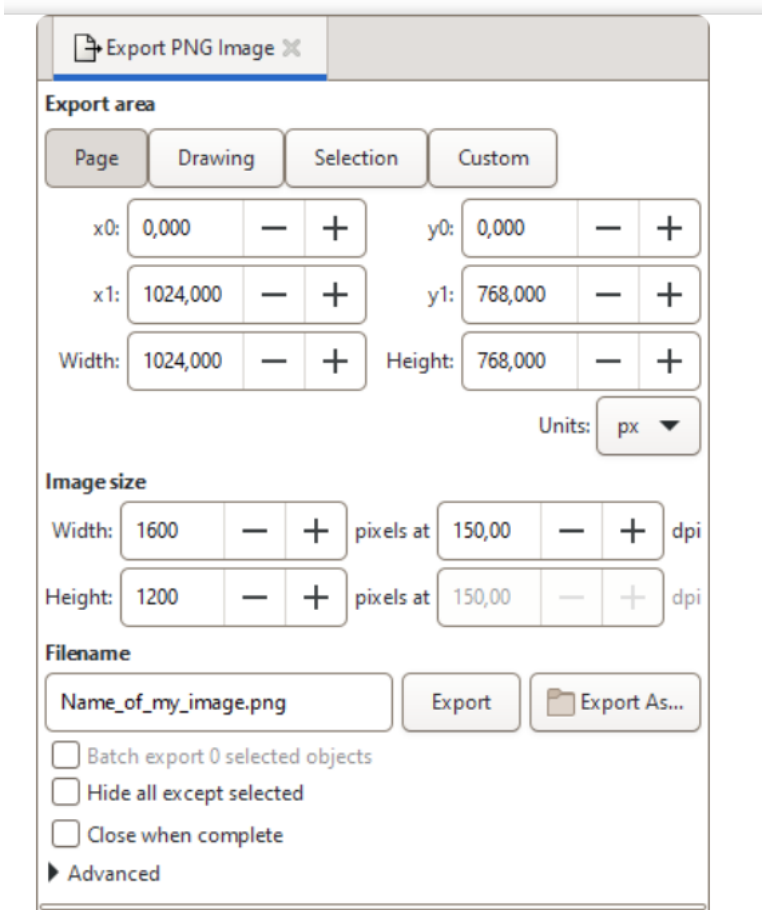
- For raster graphics file format, export to one of the following formats:
    - PNG, JPEG, TIFF

Saving a PDF

Never save as PDF only, but always keep an SVG file, because the PDF file format supports a different set of features from the SVG file format and you may lose data in the process



Exporting a PNG file



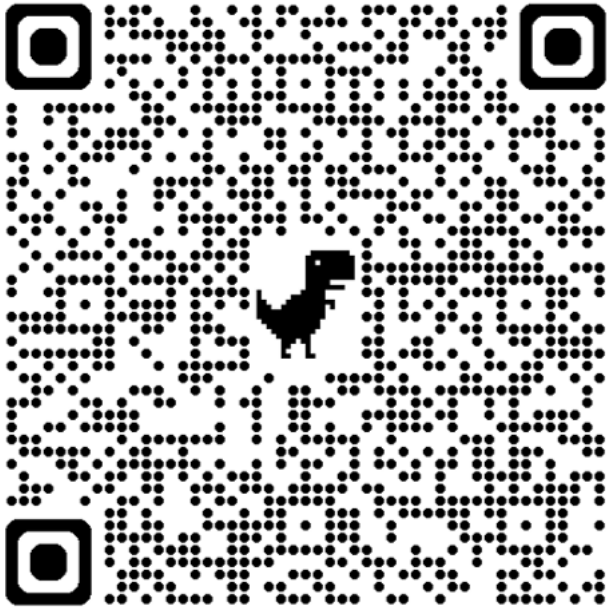
Some useful resources:

Inkscape Beginner's Guide

Inkscape manual and books

NIH BIOART source

Please share with us your feedback!



Thanks for your attention!

McGill University

Keep an eye for the workshops offered by the MiCM!

info-micm@mcgill.ca
https://www.mcgill.ca/micm/

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Canadian Centre for Computational Genomics

D2R DNA TO RNA | HeDS

References

- Inkscape manual. https://inkscape-manuals.readthedocs.io/en/latest/index.html
- Zabala A. *Designing Effective Scientific Figures: Practical Inkscape Tutorial*. Bioinformatics Core Shared Training. https://bioinformatics-core-shared-training.github.io/effective-figure-design/DesigningEffectiveScientificFigures_Practical_INKSCAPE_Zabala_v00.pdf